

CS2601 Linear and Convex Optimization

Homework 5 Solution

Due: 2021.11.4

1. Let $\bar{B} = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x}\| \leq 1\}$ be the closed unit 2-norm ball. Recall the projection of a point $\mathbf{x}_0$ onto $\bar{B}$ is the solution to the following problem,

$$\min_{\mathbf{x} \in \bar{B}} f(\mathbf{x}) = \frac{1}{2} \|\mathbf{x} - \mathbf{x}_0\|^2$$

Use the first-order optimality condition to show that the projection of $\mathbf{x}_0 \notin \bar{B}$ is $\hat{\mathbf{x}}_0 = \frac{\mathbf{x}_0}{\|\mathbf{x}_0\|}$.

Proof. Note that $\|\mathbf{x}_0\| > 1$. By the first-order optimality condition, $\hat{\mathbf{x}}_0$ is the projection $\text{proj}_{\bar{B}}(\mathbf{x}_0)$ iff

$$\begin{aligned} \langle \hat{\mathbf{x}}_0 - \mathbf{x}_0, \mathbf{x} - \hat{\mathbf{x}}_0 \rangle &\geq 0, \quad \forall \mathbf{x} \in \bar{B} \\ \iff \langle (1 - \|\mathbf{x}_0\|)\hat{\mathbf{x}}_0, \mathbf{x} - \hat{\mathbf{x}}_0 \rangle &\geq 0, \quad \forall \mathbf{x} \in \bar{B} \\ \iff \langle \hat{\mathbf{x}}_0, \mathbf{x} - \hat{\mathbf{x}}_0 \rangle &\leq 0, \quad \forall \mathbf{x} \in \bar{B} \\ \iff \langle \hat{\mathbf{x}}_0, \mathbf{x} \rangle \leq \langle \hat{\mathbf{x}}_0, \hat{\mathbf{x}}_0 \rangle &= 1, \quad \forall \mathbf{x} \in \bar{B} \end{aligned}$$

Since $\|\hat{\mathbf{x}}_0\| = 1$ and $\|\mathbf{x}\| \leq 1$ for $\mathbf{x} \in \bar{B}$, by Cauchy-Schwarz,

$$\langle \hat{\mathbf{x}}_0, \mathbf{x} \rangle \leq \|\hat{\mathbf{x}}_0\| \cdot \|\mathbf{x}\| \leq 1.$$

So the first-order optimality condition holds, and $\hat{\mathbf{x}}_0 = \text{proj}_{\bar{B}}(\mathbf{x}_0)$. □

For the rest of of this assignment, we need CVXPY (<https://www.cvxpy.org/>).

2. Consider the optimization problem

$$\begin{aligned} \min_{x_1, x_2} \quad & f(x_1, x_2) \\ \text{s.t.} \quad & 2x_1 + x_2 \geq 1 \\ & x_1 + 3x_2 \geq 1 \\ & x_1 \geq 0, x_2 \geq 0 \end{aligned}$$

Sketch the feasible set. For the objective functions in (a)-(c), first find the set of optimal solutions and the optimal value graphically, and then solve them using CVXPY. For the objective functions in (d) and (e), solve them using CVXPY. Show the outputs of CVXPY. Check the first example at <https://www.cvxpy.org/>.

org/tutorial/intro/index.html. You can use `cvxpy.maximum(x,y)` to get the maximum of two scalar variables `x` and `y`, or `cvxpy.max(x)` to get the maximum of the components of a vector variables `x`.

- (a). $f(x_1, x_2) = x_1 + x_2$
- (b). $f(x_1, x_2) = -x_1 - x_2$
- (c). $f(x_1, x_2) = x_1$
- (d). $f(x_1, x_2) = \max\{x_1, x_2\}$
- (e). $f(x_1, x_2) = x_1^2 + 9x_2^2$

Solution. The feasible set is the shaded region in Figure 1.

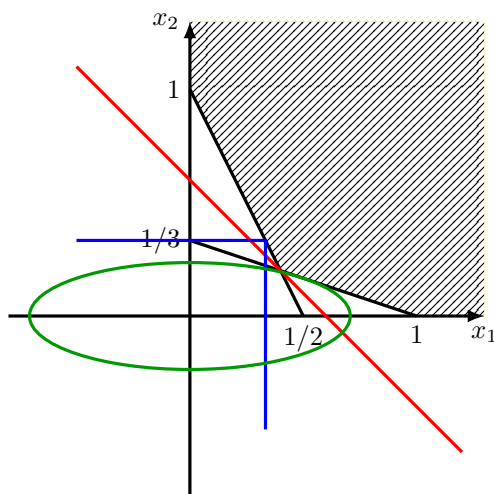


Figure 1: Problem 1

- (a). $x^* = (2/5, 1/5)$, $f^* = 3/5$. See the red line in Figure 1.

The optimal value is 0.5999999999116253
A solution x is [0.4 0.2]

- (b). No optimal solution. $f^* = -\infty$.

The optimal value is -inf
A solution x is None

- (c). $X_{opt} = \{(0, x_2) : x_2 \geq 1\}$, $f^* = 0$.

The optimal value is -2.2491441767693296e-10
A solution x is [-2.24914418e-10 1.55371590e+00]

- (d). The optimal value is 0.3333333330990559
A solution x is [0.33333333 0.33333333]

The blue curve in Figure 1 is the level set corresponding to the optimal value.

(e). The optimal value is 0.5

A solution $\mathbf{x}$ is [0.5 0.16666667]

The green curve in Figure 1 is the level set corresponding to the optimal value.

□

3. Consider the following optimization problem with $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{b} \in \mathbb{R}^m$.

$$\begin{aligned} \min_{\mathbf{x}} \quad & \|\mathbf{Ax} - \mathbf{b}\|_1 \\ \text{s.t.} \quad & \|\mathbf{x}\|_\infty \leq 1 \end{aligned} \tag{1}$$

(a). Reformulate problem (1) as an LP.

(b). Solve the original problem in (1) using CVXPY for $m = 3$, $n = 2$,

$$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & -3 \\ 1 & 2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 5 \\ 10 \\ -5 \end{pmatrix}.$$

Show the output of CVXPY. To see how to use vectors and matrices in CVXPY, check the examples at <https://www.cvxpy.org/tutorial/intro/index.html#vectors-and-matrices>. The p -norm of a vector $\mathbf{x}$ is given by `cvxpy.norm(x, p)`. Use "inf" to get the ∞ -norm. Alternatively, you can use `cvxpy.norm2(x)`, `cvxpy.norm1(x)`, `cvxpy.norm_inf(x)` to get the 2-, 1-, and ∞ -norms, respectively.

(c). Solve the LP problem you find in part (a) using CVXPY. Show the output of CVXPY. You can check the example here https://www.cvxpy.org/examples/basic/linear_program.html.

Solution. (a). Introduce auxiliary variables $\mathbf{y} \in \mathbb{R}^m$.

$$\begin{aligned} \min_{\mathbf{x}, \mathbf{y}} \quad & \mathbf{1}^T \mathbf{y} \\ \text{s.t.} \quad & -\mathbf{y} \leq \mathbf{Ax} - \mathbf{b} \leq \mathbf{y} \\ & -\mathbf{1} \leq \mathbf{x} \leq \mathbf{1} \end{aligned}$$

(b). The optimal value is 13.999999988898507

A solution $\mathbf{x}$ is [1. -1.]

(c). The optimal value is 13.999999998611605

A solution is $\mathbf{x} = [1. -1.]$, $\mathbf{y} = [4. 6. 4.]$

□

4. Let $\mathbf{w} \in \mathbb{R}^2$ and

$$\mathbf{X} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix}.$$

Solve problem (a) by hand and problems (b), (c) with CVXPY and show the outputs.

(a). Linear least squares regression

$$\min_{\mathbf{w}} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2$$

(b). Lasso

$$\begin{aligned} \min_{\mathbf{w}} \quad & \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 \\ \text{s.t.} \quad & \|\mathbf{w}\|_1 \leq t \end{aligned}$$

for $t = 1$ and $t = 10$. In each case, is the solution the same as that of (a)? Does the solution have zero components? (Note that there are numerical errors, which you should ignore when answering the questions.)

(c). Ridge regression

$$\begin{aligned} \min_{\mathbf{w}} \quad & \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 \\ \text{s.t.} \quad & \|\mathbf{w}\|_2^2 \leq t \end{aligned}$$

for $t = 1$ and $t = 100$. In each case, is the solution the same as that of (a)? Does the solution have zero components? (Again ignore the numerical errors.)

Solution. (a). The normal equation is

$$\mathbf{X}^T \mathbf{X} \mathbf{w} = \mathbf{X}^T \mathbf{y},$$

$$\text{with } \mathbf{X}^T \mathbf{X} = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}, \mathbf{X}^T \mathbf{y} = (6, 2)^T.$$

$$\mathbf{w}^* = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} = (1.5, 2)^T.$$

(b). For $t = 1$, the optimal value is 9.0000000000000002

A solution $\mathbf{w}$ is [1.00000000e+00 1.32836223e-22]

For $t = 10$, the optimal value is 4.0000000001897815

A solution $\mathbf{w}$ is [1.50000642 2.00001004]

When $t = 1$, the solution has a zero component and is different from the solution in (a). When $t = 10$, the solution is the same as that in (a) and has no zero component.

(c). For $t = 1$, the optimal value is 7.8574896437354

A solution $\mathbf{w}$ is [0.86270615 0.50570554]

For $t = 100$, the optimal value is 3.9999999993483915

A solution $\mathbf{w}$ is [1.50000161 2.00001107]

When $t = 1$, the solution is different from the solution in (a). When $t = 100$, it is the same as that in (a). In both cases, the solutions have no zero component.

□

Remark. As we discussed in class, Lasso tends to produce sparse solutions with zero components. See also Figure 2 of the Lasso paper <http://statweb.stanford.edu/~tibs/lasso/lasso.pdf>